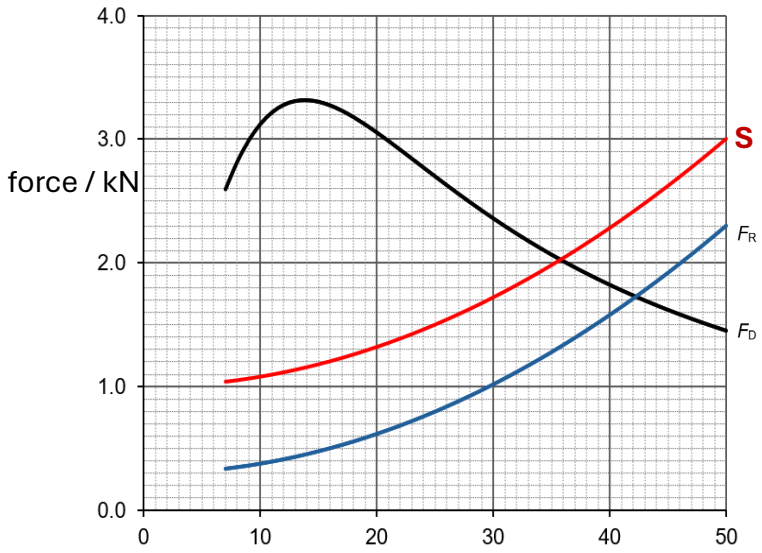
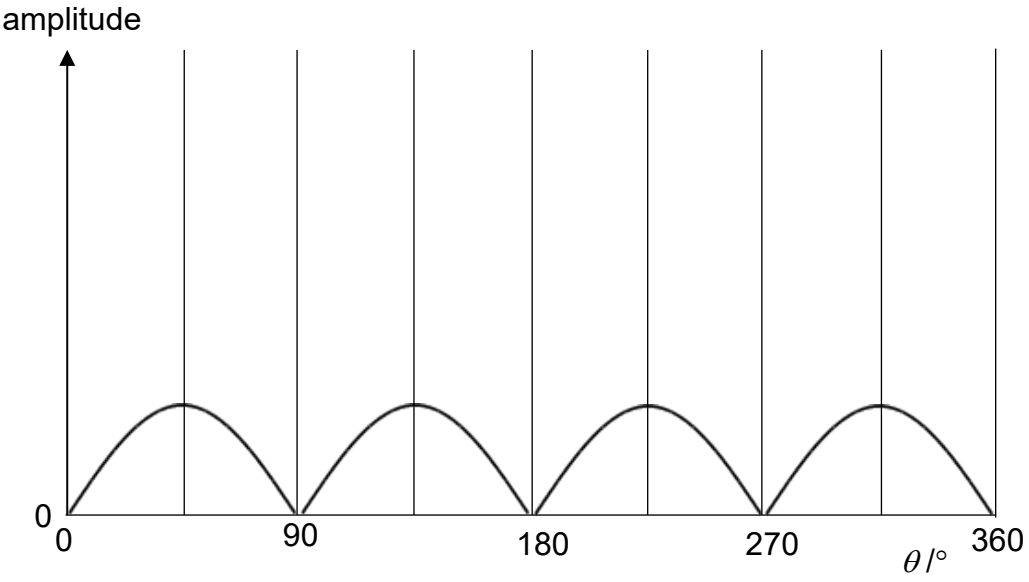
Paper 2

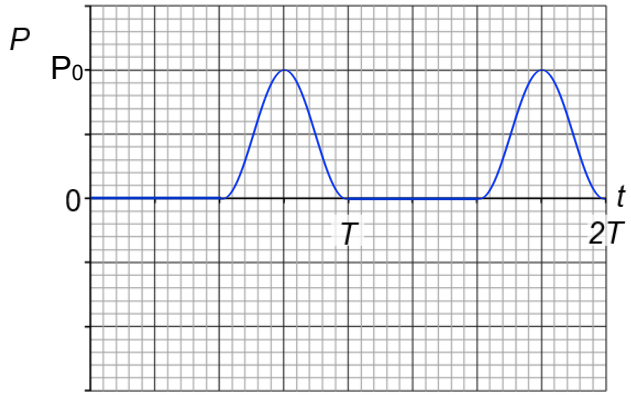
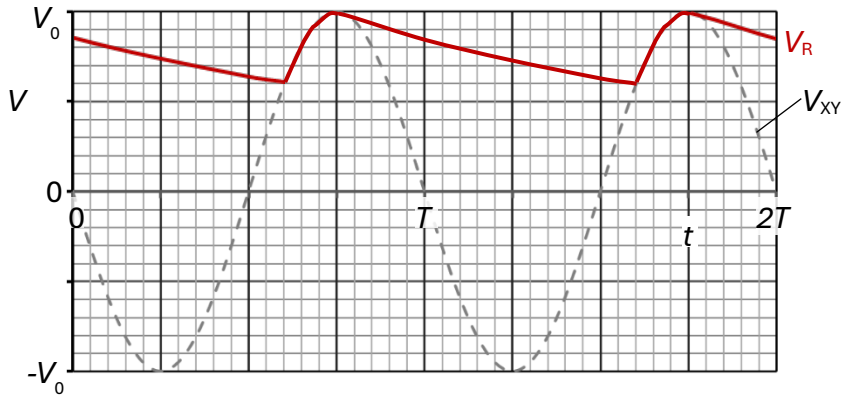
Question 1		
(a)(i)	12 to 14 m s^{-1} (The maximum acceleration occurs when the net force on the car = $F_D - F_R$ is largest.)	B1
(a)(ii)	Maximum resultant force on car, F_{net} = 3.30 kN – 0.40 kN (allow tolerance!) Maximum acceleration of the car = $\frac{(3.30-0.40)\times 10^3}{1200} = 2.42$ m s^{-2} (Allow ecf from 1(a) as long as F_{net} is > 0)	M1 A1
(a)(iii)	Rate of increase of the kinetic energy = F_{net} v = (3.30 – 0.40) × 10^3 × 12 = 3.48 x 10^4 W	C1 A1
(b)(i)	When going upslope, F_D needs to overcome both the F_R as well as the <u>component of the weight parallel to the slope.</u> Hence, the condition for <u>zero net force</u> occurs when F_D is larger and that occurs at a lower speed.	B1 B1
(b)(ii)	Correct shape – F_R shifted up by a constant value.  (The total force opposing the motion is now F_R + *mg* sin θ)	B1

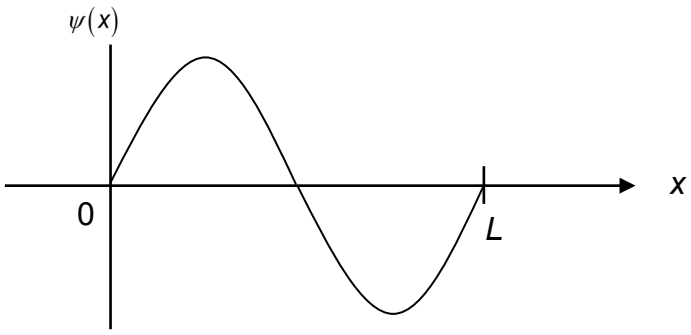
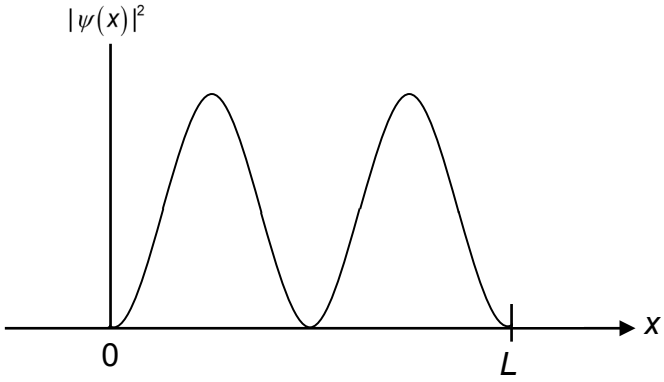
Question 2		
(a)(i)	Vertical velocity = horizonal velocity when the resultant velocity is 45° to the horizontal. $v_y = v_x = 25\cos(53°) = 15.045 = 15 \text{ m s}^{-1}$	**A1**
(a)(ii)	Taking upward as positive, $v_y = u_y + a_y t$ $15 = 25\sin(53°) + (-9.81)t$ t = 0.50 s	**A1**
(b)	$KE_{8.0} = KE_{ground} - GPE_{8.0}$ $KE_{8.0} = \frac{1}{2}(0.20)(25)^2 - (0.20)(9.81)(8.0)$ $= 47 \text{ J}$ Alternative: Accept Kinematics method.	**M1** **A1**
(c)	 **B1** – straight line with decreasing KE **B1** – non-zero KE at top of trajectory	

Question 3		
(a)	I_1 = 0.5 I_0 I_3 = 0.5 I_0 $\cos^2(25°)$ $\cos^2(65°)$ = 0.073 I_0 Note: award one mark if the student forgot to halve I_0 after passing through P_1 and gotten the final answer as 0.15 I_0.	**M1** **A1**
(b)	 **B1** – correct max and min positions **B1** – correct shape	

Question 4		
(a)	The minimas are nodes of the stationary wave / points where the incident and reflected waves meet in anti-phase / where destructive interference happen. The intensity of a wave emitted by a point source varies inversely with distance^2 / the amplitude of the wave varies inversely with distance. As plate moves rightward, the difference in intensity / amplitude between the incident and reflected waves becomes larger.	**B1** **B1** **B1**
(b)	Distance between nodes = 0.5 wavelength = 70 mm	**A1**

Question 5		
(a)	Electric field between plates M and N $E = \frac{V}{d} = \frac{235-(-235)}{2.00\times10^{-2}} = 2.35\times10^{4}\ \text{V m}^{-1}$ $F = Eq = \left(2.35\times10^{4}\right)\left(58\times10^{-12}\right) = 1.4\times10^{-6}\ \text{N}$	**B1** **A1**
(b)	Acceleration, $a = \frac{F}{m} = \frac{1.363\times10^{-6}}{0.12\times10^{-3}} = 1.136\times10^{-2} = 1.1\times10^{-2}\ \text{m s}^{-2}$	**A1**
(c)	Time to travel from M to N $s = ut + \frac{1}{2}at^2$ $2.0\times10^{-2} = 0 + \frac{1}{2}\left(1.136\times10^{-2}\right)\left(t^2\right)$ $t = 1.8765\ s$ Frequency, $f = \frac{1}{2\times t} = \frac{1}{2\times1.8765} = 0.266\ \text{Hz}$	**B1** **B1** **A1**
(d)	The acceleration for simple harmonic motion is proportional to its displacement. Since the acceleration of the sphere is constant, the motion cannot be simple harmonic.	**A1**

Question 6		
(a)(i)	Two sections of zero power of width $0.5T$ starting at $t = 0$ and $t = T$ Two humps of width $0.5T$ and peak P_0 *starting* at $t = 0.5T$ and $t = 1.5T$ [Deduct only this mark if the positions of the zero power and humps are swapped.] Correct sinusoidal-squared shape for the 2 non-zero humps (i.e. with smooth troughs seating on the t-axis at $P = 0$ 	**B1** **B1**
(a)(ii)	The mean power within each hump is $\frac{1}{2}P_0$ from the symmetry of the curve. An additional half factor is from the removal of half of the power in each cycle (resulting in the ¼ factor).	**B1** **B1**
(a)(iii)	$\frac{V_{OUT,rms}^{\ 2}}{R} = \frac{1}{4}P_0 = \frac{1}{4}(2\frac{V_{XY,rms}^2}{R})$ $V_{OUT,rms} = \frac{V_{XY,rms}}{\sqrt{2}} = \frac{12}{\sqrt{2}} = 8.485 = 8.5\ \text{V}$	**M1** **A1**
(b)(i)	Correct shape.  Note: No penalty if the first part of the graph is not drawn.	**B1**
(b)(ii)	From the graph, the time for the capacitor to discharge from V_0 to 0.60 V_0 is $0.85T$. $V = V_0 e^{-\frac{t}{\tau}} \Rightarrow \tau = \frac{-0.85T}{\ln 0.60} = 1.7T$	**B1** **A1**

Question 7		
(a)(i)	 Or negative sine.	**B1**
(a)(ii)		**B1**
(b)(i)	A normalised wave function is one where the probability of finding the particle in all space / across all x equals 1 Or where the total area under the probability density function equals 1.	**B1**
(b)(ii)	$\frac{A^2}{2} \times L = 1$ $A = \sqrt{\frac{2}{L}}$	**B1**
(c)	Energy of photon emitted $= \frac{(2)^2 h^2}{8mL^2} - \frac{h^2}{8mL^2} = \frac{3h^2}{8mL^2}$ $= \frac{3\left(6.63 \times 10^{-34}\right)^2}{8\left(0.15 \times 9.11 \times 10^{-31}\right)\left(2.2 \times 10^{-9}\right)^2}$ $= 2.4923 \times 10^{-19}$ J Wavelength of photon emitted $= \frac{hc}{E}$ $= \frac{\left(6.63 \times 10^{-34}\right)\left(3.00 \times 10^{8}\right)}{2.4923 \times 10^{-19}}$ $= 7.98 \times 10^{-7}$ m	**C1** **C1** **A1**

Question 8		
(a)	Ther rate of decay is not affected by external factors like temperature and pressure nor by external stimuli.	**B1**
(b)(i)	Alpha radiation is unsuitable because its range in human tissue is too short, meaning it cannot penetrate deeply enough to destroy the entire tumour mass.	**B1**
(b)(ii)1.	Number of iodine particles = $\frac{1.2\times10^{-9}}{131}\times6.02\times10^{23}=5.5\times10^{12}$ Activity = λN Activity = $\frac{8.6\times10^{-2}}{24\times3600}\times(5.5\times10^{12})$ $=5.5\times10^{6}$ Bq	**C1** **C1** **A1**
(b)(ii)2.	$A=A_o e^{-\lambda t}$ $\frac{1}{50}=e^{-\frac{8.6\times10^{-2}}{24\times3600}t}$ $t=3.9\times10^{6}$ s	 **C1** **A1**

Question 9		
(a)	Stage 2: ${}_1^2\mathrm{H} + {}_1^1\mathrm{H} \rightarrow {}_2^3\mathrm{He} + \gamma$ Stage 3: ${}_2^3\mathrm{He} + {}_2^3\mathrm{He} \rightarrow {}_2^4\mathrm{He} + 2{}_1^1\mathrm{H}$ Accept ${}_1^1p$ for proton.	**A1** **A1**
(b)(i)	$E_K = \frac{3}{2}kT = \frac{3}{2} \times 1.38 \times 10^{-23} \times (1.5 \times 10^7 + 273.15) = 3.105 \times 10^{-16} = 3.1 \times 10^{-16}$ J Student need to either show that they convert the temperature to kelvin or explain why they didn't convert to get the marks	**A1**
(ii)	Applying conservation of energy to determine the minimum distance the 2 protons can approach each other: Loss in kinetic energies of 2 protons = Gain in electric potential energy of protons $2E_K - 0 = \frac{e^2}{4\pi\varepsilon_0 x}$ $x = \frac{e^2}{8\pi\varepsilon_0 E_K}$ $= \frac{\left(1.60 \times 10^{-19}\right)^2}{8\pi\varepsilon_0 \left(3.1 \times 10^{-16}\right)}$ $= 3.7 \times 10^{-13}$ m Since d is larger than 10^{-14} m, the protons are not close enough to fuse.	**C1** **A1** **A1**
(c)(i)	The wave function represents the quantum state of a particle, providing all its information such as its position, momentum, etc.	**A1**
(c)(ii)1.	Since the amplitude of wave function after the barrier is non-zero, the probability density function, given by the square of the wave function, also non-zero, implies there is a possibility of finding the particle in this region.	**B1** **B1**
(c)(ii)2.	The wavelength of the wavefunction remains the same, from de Broglie's relationship, the momentum and hence energy of the particle remains the same after passing through the barrier.	**A1** **B1**

(d)(i)	Along the horizontal axis: Magnetic force provides the centripetal force $Bqv_x = mr\omega^2$ $Bqv_x = mv_x\omega$ $\frac{2\pi}{T} = \frac{Be}{m}$ $T = \frac{2\pi \times 9.11\times10^{-31}}{40\times10^{-6}\times1.60\times10^{-19}}$ $= 8.94\times10^{-7}$ s Along the vertical axis: $v_y = 5.0\times10^5 \cos 25^o$ $d = v_y T = (5.0\times10^5 \cos 25^o)\times 8.94\times10^{-7} = 0.41$ m	**B1** **B1** **B1** **A1**
(d)(ii)	Change: Opposite direction. Explain: Proton is positive. Change : Radius will increase. Explain: Proton is more massive. Any other reasonable answer.	**B1** **B1** **B1** **B1**
(d)(iii)	Charged particles in the solar transfer energy to the atoms (or electrons in the atoms) during collisions, exciting them to higher energy levels. When excited atoms/electrons in atom de-excite, photons are emitted.	**A1** **A1**
(e)(i)	Intensity $= \frac{P_{sun}}{4\pi R^2} = \sigma T^4$ $T = \left(\frac{4.0\times10^{26}}{4\pi\times(1.4\times10^{10})^2\times5.7\times10^{-8}}\right)^{1/4}$ $= 1299 = 1300$ K	**B1** **C1** **A1**
(e)(ii)	Help forecast of solar flares, changes in solar wind as modern world depend on devices are sensitive to electromagnetic variations (eg navigation systems, high voltage power grid)	**A1**